AI/ML Engineering Roadmap

A Complete Self-Study Path from Foundations to Professional AI Engineering

Study Order

Phase 0	Math Foundations + Python (run in parallel throughout)
Phase 1	Python · Statistics · EDA · Visualization (GFG Weeks 0–4)
Phase 2	SQL + Databases
Phase 3	Machine Learning + WQU Data Science Projects (GFG Weeks 5–10)
Phase 4	Git · Docker · CI/CD · Core MLOps
Phase 5	Deep Learning+ WQU DL Projects + Reinforcement Learning (GFG Weeks 11–16)
Phase 6	Modern AI Engineering (NLP · CV · LLMs · GenAI · RAG · Vector DBs · Agents)
Phase 7	Advanced Track — Cloud · Data Engineering · Google Professional ML Engineer

PHASE 0 Python + Math Foundations

Run Python (0.0) alongside every phase. Complete Math (0.1–0.4) before starting Phase 1.

0.0 — Python

→ **Python Programming MOOC 2026** — *University of Helsinki · free · world-class*
https://programming-26.mooc.fi/

0.1 — Calculus

→ **AP Calculus AB** — *Khan Academy*
https://www.khanacademy.org/math/ap-calculus-ab

0.2 — Linear Algebra

→ **Linear Algebra** — *Khan Academy*
https://www.khanacademy.org/math/linear-algebra

0.3 — Statistics & Probability

→ **Statistics & Probability** — *Khan Academy*
https://www.khanacademy.org/math/statistics-probability

Alternative Path — Mathematics for Machine Learning (if Khan Academy feels too disconnected from AI)

Mathematics for Machine Learning Specialisation — *Imperial College London · Coursera · audit free · Linear Algebra, Multivariate Calculus & PCA*

0.1 — Calculus

→ **Multivariate Calculus** — *Imperial College London*
https://www.youtube.com/watch?v=cWZLPv4ZJhE&list=PLiiljHvN6z193BBzS0Ln8NnqQmzimTW23

→ **Essence of Calculus** — *3Blue1Brown · watch first*
https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr

0.2 — Linear Algebra

→ **Linear Algebra for Neural Networks** — *Imperial College London*
https://www.youtube.com/playlist?list=PLiiljHvN6z1_o1ztXTKWPrShrMrBLo5P3

→ **Essence of Linear Algebra** — *3Blue1Brown · must watch*
https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

— PCA (after linear algebra)

→ **PCA Course** — *Marc Peter Deisenroth · co-author of Mathematics for Machine Learning*
https://www.youtube.com/playlist?list=PL93aLKqThq4iDbY_wxfh3WgxkkzqH4aoo

→ **Companion Textbook (free PDF)** — *mml-book.com*
https://mml-book.com

PHASE 1 Python · Statistics · EDA · Visualization

✅ Fully covered by GFG Weeks 0–4. No external course required.

1.1 Data Science Overview — *GFG Week 0*
1.2 Python Foundations — *GFG Week 1*
1.3 Statistics for Data Science — *GFG Week 2*
1.4 EDA with Pandas & NumPy — *GFG Week 3*
1.5 Data Visualization — Matplotlib, Seaborn, Plotly — *GFG Week 4*

GFG Course: https://www.geeksforgeeks.org/batch/ds-16?tab=Resources

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PHASE 2 SQL + Databases

⚠ GFG does not teach SQL. Complete the resources below before Phase 3.

2.1 SQL Murder Mystery — *Joon Park and Cathy He*

https://www.kaggle.com/learn/intro-to-sql

2.2 SQL Tutorial 1 — *Mode Analytics · primary*
https://mode.com/sql-tutorial/

2.3 SQL Tutorial 2 — *SQL Zoo*
https://sqlzoo.net/wiki/SQL_Tutorial

2.4 PostgreSQL Basics — *YouTube · optional but recommended*
https://www.youtube.com/watch?v=qusjTv8N1cA

PHASE 3 Machine Learning + WQU DS Projects

✅ Fully covered by GFG Weeks 5–10. No external course required.

3.1 ML Basics + Regression — *GFG Week 5*

3.2 Model Evaluation + SHAP / Model Explainability + Classification — *GFG Week 6 + SHAP*

Do SHAP / Model Explainability immediately after Day 36. It takes one afternoon and you'll use it in every project after.

3.2.1 SHAP / Model Explainability — https://shap.readthedocs.io/en/latest/

3.3 ML Algorithms — Decision Trees, SVM, KNN, Naive Bayes — *GFG Week 7*

3.4 Ensemble Learning — XGBoost, LightGBM, CatBoost — *GFG Week 8*

3.5 Unsupervised Learning — Clustering, GMM, DBSCAN — *GFG Week 9*

3.6 Dimensionality Reduction + Deployment Introduction — *GFG Week 10*

3.7 — WQU Data Science Projects

→ **Applied Data Science Lab**
https://www.wqu.edu/adsl

PHASE 4 Git · Docker · CI/CD · Core MLOps

⚠ GFG does not teach these tools. Complete before moving to Deep Learning.

4.1 — Git & GitHub

→ **Git & GitHub Full Course** — *freeCodeCamp*
https://www.freecodecamp.org/news/git-and-github-crash-course-for-beginners/

→ **Microsoft Learn for GitHub** — *Microsoft*
https://learn.microsoft.com/en-us/training/github/

4.2 — Docker

→ **Docker Tutorial for Beginners** — *TechWorld with Nana · YouTube*
https://www.youtube.com/watch?v=3c-iBn73dDE

→ **Docker for Beginners** — *freeCodeCamp*
https://www.freecodecamp.org/news/an-introduction-to-docker-and-containers-for-beginners/

4.3 — CI/CD

→ **GitHub Actions Full Course** — *YouTube*
https://www.youtube.com/watch?v=mFFXuXjVgkU

4.4 — Core MLOps

→ **Made With ML — MLOps** — *madewithml.com · best free MLOps resource available · primary*
https://madewithml.com

→ **MLOps Zoomcamp** — *DataTalksClub · alternative / highly practical*
https://datatalks.club/blog/mlops-zoomcamp.html

PHASE 5 Deep Learning + WQU DL Projects + Reinforcement Learning + DSA

✅ Fully covered by GFG Weeks 11–16. No external course required.

5.1 Introduction to Deep Learning — *GFG Week 11*
5.2 Optimisation + MLP + Backpropagation — *GFG Week 12*
5.3 CNN + RNN + Time Series — *GFG Week 13*
5.4 LSTM, GRU, Transformers — *GFG Week 14*
5.5 GANs + Autoencoders — *GFG Week 15*

✅ GFG backup

5.6 Practical Deep Learning ***fast.ai***
https://course.fast.ai/

5.5 Dive into Deep Learning
https://d2l.ai/

5.6 — WQU Deep Learning Projects

→ **Deep Learning Fundamentals Lab**
https://www.wqu.edu/deep-learning-lab

5.7— Core Reinforcement Learning

5.7.1 Reinforcement Learning — *GFG Week 16*

5.7.2 Reinforcement Learning — ***Hugging Face Deep RL Course***
https://huggingface.co/learn/deep-rl-course/unit0/introduction

5.7.2 PPO (Proximal Policy Optimization) — ***Hugging Face Deep RL Course***
https://huggingface.co/learn/deep-rl-course/unit8/introduction

5.8 — DSA

5.8.1 Problem Solving with Algorithms and Data Structures using Python — *Runestone Academy*
https://runestone.academy/ns/books/published/pythonds/index.html

5.8.2 DSA *(Video lectures)* — *freeCodeCamp*
https://www.youtube.com/watch?v=pkYVOmU3MgA

5.8.3 Practice Problems — *Striver's A2Z Sheet*
https://takeuforward.org/dsa/strivers-a2z-sheet-learn-dsa-a-to-z

PHASE 6 Modern AI Engineering

📖 **Companion Document**

Phase 6 has a dedicated full course guide covering all topics below in depth, with the top 3 free comprehensive courses per subject, structured in the exact order you should study them.

→ **See: Modern AI Engineering — Full Course Guide**

Topics — in learning order

6.1 Natural Language Processing — *Transformers, BERT, GPT, text classification, NER, summarisation*
6.2 Computer Vision — *CNNs, object detection, segmentation, OpenCV, real-time applications*
6.3 Large Language Models (LLMs) — *Architecture, training dynamics, fine-tuning, deployment*
6.4 Generative AI — *RLHF, diffusion models, full-stack GenAI engineering*
6.5 RAG — Retrieval-Augmented Generation — *Building, evaluating, and optimising RAG pipelines*
6.6 Vector Databases — *Embeddings, ANN search, ChromaDB, Weaviate, Pinecone, hybrid search*
6.7 Agentic AI — *Agents, tool use, multi-agent systems, LangGraph, AutoGen*
6.8 Prompt Engineering + Fine-Tuning — *Prompt design, LoRA, QLoRA, PEFT, instruction tuning*
6.9 FastAPI for AI Services — *Building and deploying AI-powered APIs*

PHASE 7 Advanced Track — Cloud · Data Engineering · Certification

📌 Complete Phases 5 and 6 before starting Phase 7. These topics require full deep learning and AI engineering knowledge to use meaningfully.

7.1 — Data Engineering

→ **Data Engineering Zoomcamp** — *DataTalksClub · free · Kafka, Spark, dbt, BigQuery, Airflow*
https://datatalks.club/blog/data-engineering-zoomcamp.html

7.2 — Google Cloud + ML on Cloud

→ **Machine Learning Path** — *Google Cloud Skills Boost · official & free · Vertex AI, BigQuery ML, Pipelines*
https://www.cloudskillsboost.google/paths/17

7.3 — Advanced MLOps at Scale

→ **MLOps Zoomcamp** — *DataTalksClub · complete if not done in Phase 4*
https://datatalks.club/blog/mlops-zoomcamp.html

7.4 — Google Professional Machine Learning Engineer ★ Capstone

The Google Professional ML Engineer is one of the most respected cloud ML credentials globally. Attempt only after completing all phases above.

Step 1 **Google Cloud Skills Boost — ML Path** — *Official pre-certification curriculum — complete first*
https://www.cloudskillsboost.google/paths/17

Step 2 **Official Exam Guide** — *Free · everything testable is listed here*
https://cloud.google.com/learn/certification/machine-learning-engineer

Step 3 **Preparing for the Professional ML Engineer Exam** — *Google + Coursera · audit free · official exam prep*
https://www.coursera.org/learn/preparing-cloud-professional-ml-engineer-exam

Step 4 **Hands-on Vertex AI Labs** — *Google Cloud Skills Boost · complete every ML quest*
https://www.cloudskillsboost.google/

GeeksForGeeks Course (all GFG phases): https://www.geeksforgeeks.org/batch/ds-16?tab=Resources

DeepLearning.AI Short Courses (Phase 6 permanent resource): deeplearning.ai/short-courses

GPU Access — Kaggle Notebooks (Free T4 GPU · 30hrs/week · use for Phase 5, 6.8, and ARENA): www.kaggle.com/code

GPU Access — Google Colab (Free T4 GPU backup · use when Kaggle quota runs out ·): https://colab.research.google.com/